



Fig. 2: Process of ultrasonic cleaning



Fig. 3: Types of ultrasonic cleaners



Fig. 4: Ultrasonic cleaning solutions



Fig. 5: Electronics of a Chinese ultrasonic cleaner

steel tank (usually accompanied by a steel mesh basket to keep the object), also known as the ultrasonic tank.

A transducer is usually placed at the bottom or on the sides, or sometimes both when watt density is a concern. It can be welded directly into the tank, or a watertight immersible unit can be placed directly into the aqueous solution. The ultrasonic generator converts a standard electrical frequency of 50Hz - 60Hz into high frequency required for the task, in the ultrasonic range.

A piezoelectric transducer, commonly used in common models, contains piezo-ceramic crystals that change their size and shape when exposed to ultrasonic waves. This causes the sides and bottom of the tank to vibrate and form innumerable microscopic bubbles in the fluid. As

these expand and compress alternately, a vacuum cavity builds up within these.

In addition, the crystals' constant impact against each other produces heat energy. Since cleaning of some objects calls for precise time and temperature control, a timer and heater circuitry is also included in the ultrasonic cleaner. (Heated ultrasonic cleaning improves the cleaning efficiency by removing trapped air in the fluid.)

Besides, auto-follow electronic circuitry in latest ultrasonic generators is designed to maintain the centre frequency when the ultrasonic tank is subjected to varying load conditions. When parts to be cleaned are placed in the tank, or when water level changes, the load on the generator changes. With auto-follow circuitry, the generator matches electrically with the mechanical load, providing optimum output at all times to the ultrasonic tank.

When it comes to the cleaning solution, regular tap water works just fine for a desktop (home-grade)

ultrasonic cleaner. Electronic goods and jewellery, however, need a more powerful cleaning agent. Note that, water-soluble special ultrasonic cleaning solution is also available. It conducts ultrasonic waves better than regular water. Some of these ultrasonic cleaning solutions are general-purpose, while others are intended for specific objects.

Inside electronics

Electronics of an ultrasonic cleaner consists of a power supply, an ultrasonic generator and transducers. Usually, a main printed wired board (PWB) holds all components of the power supply and the ultrasonic generator. A sub-circuit board is used for the microcontroller based timer/heater commander, and this board also carries user interface keys, indicators, the display panel and so on. Whole electronics of a Chinese ultrasonic cleaner (40kHz/100W) is shown in Fig. 5.

Piezoelectric ultrasonic transducers are made up of several components. The ceramic (usually lead zirconate) crystal is sandwiched between two strips of tin. When sufficient voltage is applied across the strips, it creates a displacement in the crystal, known as piezoelectric effect. In principle, the ultrasonic transducer is effectively a series-resonant LC circuit, where piezo plate electrodes form a capacitor, and the resonating mass acts as an inductance.

The metal horn attached to the transducer focuses the energy into a small area, as demanded by the specific application. The horn vibrates at its own resonant frequency, so resonant frequency of the complete setup (transducer and horn) is a blend of the two. When the transducers are epoxied to a diaphragm on the walls or the bottom of the tank, displacement in the crystal causes a movement of the diaphragm, which, in turn, causes a pressure wave to be transmitted through the aqueous solution in the tank.

The transponder driver circuit is, in fact, a variation of the famed Royer